

# LLM and Graphs for Financial Texts

A Cross-Semester Project Report for the European Investment Bank

IEOR 4737 — AI Applications in Finance  
Columbia University — EIB 1  
Fall 2025 • Spring 2026 • Summer 2026

August 19, 2026

## Abstract

This report documents the full lineage of the EIB Knowledge Graph project across three consecutive Columbia University teams (Fall 2025, Spring 2026, Summer 2026). It covers the data, the extraction and evaluation pipeline, the graph neural network used for link prediction, the recent multi-model LLM benchmark, and the interactive web application built this summer.

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# 1 Executive summary

- **Business question.** How to turn the daily firehose of financial news into a structured, searchable view of what companies, sectors, and instruments are affecting each other over time.
- **Approach.** Large language models (LLMs) read each news article and extract *triplets* — short factual statements of the form “*Nvidia* — *competes with* — *AMD*”. Triplets accumulate into a knowledge graph. A graph neural network trained on the graph then predicts which entity relationships are likely to appear next.
- **Three semesters of work.**
  - Fall 2025 built the first end-to-end pipeline on the FNSPID financial news dataset.
  - Spring 2026 hardened the pipeline: fixed a data-leakage bug in the evaluation, showed full article text beats article summary for extraction, added causal-feature augmentation that stabilised the model through Covid-19.
  - Summer 2026 (current) added: multi-LLM benchmark (Claude, GPT, Qwen, Gemini) with automated quality scoring; a public, deployed web application; a FinDKG-style predictions leaderboard driven by real GAT weights.
- **Current state.**
  - Live web application at <https://eib-knowledge-graph.vercel.app>.
  - 24 months of knowledge graph snapshots (2019-01 through 2020-12),  $\approx$  12,000 triplets,  $\approx$  8,000 entities.
  - GAT predictions layer with MRR 0.77 on rolling-window link prediction.
  - Multi-model LLM benchmark completed for four models on identical inputs, scored on four quality dimensions by an independent Judge LLM.

## 2 Project timeline

### 2.1 Fall 2025 — Baseline pipeline

*Lee, G.; Gao, J.; Lu, X.; Cuadros, A.*

- Ingested the FNSPID financial news dataset (Zihan1004 on Hugging Face).
- Restricted focus to semiconductor-related news (2020–2024).
- Extracted triplets using a locally-hosted Qwen 2.5 LLM via Ollama.
- Reverse-engineered the four Judge-LLM quality criteria into the initial extraction prompt itself — shortcutting a slow generate-then-critique loop.
- Defined the four quality dimensions: Coverage, Uniqueness, Semantic Validity, Consistency.
- Built the first Graph Attention Network (GAT) for link prediction; trialled five loss functions (BPR, InfoNCE, Cross-Entropy, FindKG, Hybrid).

- Defined three graph-construction strategies keyed to Judge output: *Original*, *Fallback*, *Strict*.
- Delivered an initial static HTML visualisation of monthly knowledge graphs.

## 2.2 Spring 2026 — Robustness and causality

*Xixi Chen; Xinyi Chen; Yi Yang.*

- Systematically compared triplet extraction from *full article text* versus *article summary*. Full text won by  $\approx 0.08$  MRR — coverage dominates redundancy.
- Identified and fixed a temporal data-leakage bug in the rolling-window evaluation: the original code built its global entity index from the full dataset before splitting into train/test, so the model implicitly saw future entities through its embedding table.
- Ran a window-size ablation: 3-month training window beat 6-month and 12-month across every configuration — recency dominates historical depth in a non-stationary market.
- Added *causal feature augmentation*: encoded each entity’s causal role (in-degree, out-degree, PageRank on a NOTEARS-learned DAG) as a 5-dimensional add-on to the 33-dimensional baseline features.
- Case-studied two market events: Covid-19 (gradual, 2019–2020) and the Russia–Ukraine war (sudden, 2021–2022).
  - Covid-19: causal features cut the shock-period MRR drop by 96%.
  - Ukraine war: absolute performance higher; within-shock variance dramatically lower.
- Consolidated the pipeline into a documented package with a Rolling Window subfolder for the corrected training code.

## 2.3 Summer 2026 — Comparison, UI, deployment

*Alexandra Paiz; Pierre Pujol (GitHub [SigmaWave](#)); Ruiwen Wang.*

- Ran a controlled multi-model LLM benchmark on identical inputs: Claude Sonnet 5, GPT-5, Qwen 2.5 14B, Gemini 2.5 Flash. All extractions scored by the same Judge LLM (Qwen 2.5 14B via Ollama) on Coverage, Uniqueness, Semantic Validity, Consistency.
- Retrained the GAT on the 2019–2020 corpus with the Spring 2026 rolling-window fix; produced 230 checkpoints (10 model configurations  $\times$  23 monthly validation periods).
- Built an interactive, publicly deployed web application (Next.js 16, Cytoscape.js, Tailwind CSS, shadcn/ui) at <https://eib-knowledge-graph.vercel.app>.
- Added a FinDKG-style “Top KG Entities” predictions leaderboard driven by real GAT embeddings, per month, exposing rank percentile, novelty z-score, 3-month trend sparkline, and top predicted-impacted financial entities.
- Deployed a self-updating news scraper framework (Google News RSS + regulator RSS feeds; GitHub Actions cron; per-corpus watchlists).

- Deployed static builds to Vercel with automatic redeploy on every push to `main`.
- Stood up a fully automated **live corpus** — STOXX Europe 600, 30 curated tickers, refreshed daily by GitHub Actions with zero manual steps (Section 11).
- Built a **news factor model** (five factors + PCA + KMeans archetype clustering) over the live corpus, exposed as a dedicated UI tab (Section 13).
- Brought the extraction up to the three-LLM standard of the team pipeline: judge-and-refine pass plus cross-article entity canonicalisation (Section 12).
- Built the **VM-to-UI publishing pipeline** and used it to ship the full 2-year Qwen extraction (5,358 articles / 30,861 triplets) to the public UI (Section 14).

## 3 Data

### 3.1 Source corpus

- **FNSPID** (Financial News and Stock Price Integration Dataset), Hugging Face Zihan1004/FNSPID.
- 15.7M financial news articles, 1999–2023, mapped to NASDAQ tickers.
- Full articles include title, body, ticker, timestamp, URL.

### 3.2 Working subset for this project

- **Semiconductor slice** used throughout:  $\approx 2,300$  articles across 2019–2020, chosen for its coverage of a full pre-shock / shock / post-shock cycle (Covid-19).
- Second slice used for the Spring 2026 causal study:  $\approx 2021$ –2022, spanning the Russia–Ukraine invasion.

### 3.3 Extracted knowledge graph

- File: `summary_triplets_19_20.csv` (Spring 2026 deliverable).
- $\approx 11,800$  triplets after quality filtering.
- $\approx 8,000$  unique entities across the 24 monthly snapshots.
- Entity types normalised to a closed vocabulary of 18 categories (company, stock ticker, sector, financial instrument, macro indicator, etc.).

## 4 Pipeline overview

Four sequential stages, unchanged in intent since Fall 2025, refined in every semester:

1. **Triplet extraction.** LLM reads each news article, returns a list of (entity<sub>1</sub>, type<sub>1</sub>, relation, relation category, tuples).
2. **Quality evaluation.** An independent Judge LLM scores each extraction on four dimensions and, if flagged, requests a revision.

3. **Graph construction.** Triplets become edges of a directed graph; entities become nodes. One graph per calendar month.
4. **Link prediction.** A Graph Attention Network is trained on trailing months and evaluated on the next month — the “predictable” relationships become the model’s output.

## 5 Stage 1 — Triplet extraction

### 5.1 The triplet as unit of knowledge

- A triplet is one factual claim: “*Taiwan Semiconductor* — *manufactures for* — *AMD*”.
- Six-slot format:  $(e_1, \text{type}_1, r, c, e_2, \text{type}_2)$  — two entities, their types, the relationship verb, and the relationship’s high-level category.
- Relationship categories collapse the hundreds of verbs into three families:
  - *SSI* — Sector-Specific Insights (product launches, R&D, deals).
  - *GMM* — Global Market Movements (macro drivers, cross-market flows).
  - *GFMK* — General Financial Market Knowledge (ownership, ticker mapping, structural facts).
- Rationale: keep the vocabulary permissive at the verb level while giving downstream models a coarse handle on relationship meaning.

### 5.2 Prompt engineering

- Prompt template lives in `components/tgp.txt` (triplet generation prompt).
- Instructs the model to: identify entities, classify each entity type against the ontology, propose the relationship verb, tag the category, avoid duplicates.
- The Fall 2025 team pre-loaded the four Judge criteria into this prompt — so the extractor already tries to satisfy Coverage / Uniqueness / Validity / Consistency in a single pass.

### 5.3 Models used across semesters

- **Fall 2025:** Qwen 2.5 (14B and 32B variants) locally via Ollama.
- **Spring 2026:** Same Qwen 2.5, focus was downstream not on the extractor.
- **Summer 2026:** Multi-model comparison — Qwen 2.5 14B (baseline), Claude Sonnet 5, GPT-5, Gemini 2.5 Flash.

### 5.4 Multi-LLM benchmark (Summer 2026)

- **Design.** Same 127-article slice of the corpus fed to each model; identical prompt; identical Judge LLM (Qwen 2.5 14B) scoring the outputs. Fully apples-to-apples.
- **Cost + speed.**

- Gemini 2.5 Flash: 129 articles, 1,213 triplets,  $\approx \$0.05$ , 18.7s / article average. 100% well-formed output.
- Claude / GPT / Qwen: costs and speeds recorded in Pierre’s benchmark deck.
- **Robustness.** Signal.SIGALRM per-article timeout (60s) added to prevent silent hangs; incremental CSV checkpointing every 10 articles so an aborted run resumes cheaply.
- **Results** (see Section 6.4) — Gemini 2.5 Flash’s Judge scores: Coverage 0.74, Uniqueness 0.83, Semantic Validity 0.84, Consistency 0.92 on 127 articles.

## 6 Stage 2 — Judge LLM quality evaluation

### 6.1 Motivation

- Raw LLM output is noisy: hallucinated entities, mis-typed entities, redundant triplets, contradictions.
- Feeding noise into the graph pollutes downstream training.
- Solution: a second LLM (the “Judge”) re-reads each extraction and scores it.
- Judge is independent from the extractor to avoid confirmation bias.

### 6.2 The four dimensions

**Coverage** — fraction of salient content in the source article captured by the triplet set.

$$\text{Coverage}(a) = \frac{|\{k \in \mathcal{K}_a \mid k \text{ referenced by some } t_i \in \mathcal{T}_a\}|}{|\mathcal{K}_a|}$$

where  $\mathcal{K}_a$  is the set of salient semantic units in article  $a$  and  $\mathcal{T}_a$  is the extracted triplet set.

**Uniqueness** — penalises redundant triplets (measured via cosine similarity in embedding space):

$$\text{Uniqueness}(a) = 1 - \frac{2}{n(n-1)} \sum_{i < j} \mathbb{I}(\cos(\phi(t_i), \phi(t_j)) > \tau)$$

**Semantic Validity** — fraction of triplets that are semantically well-formed against the financial ontology:

$$\text{Validity}(a) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(t_i \text{ is semantically valid})$$

**Consistency** — absence of pairwise contradictions:

$$\text{Consistency}(a) = 1 - \frac{|\{(t_i, t_j) \mid t_i \perp t_j\}|}{\binom{n}{2}}$$

All four are in  $[0, 1]$ ; higher is better.

### 6.3 Three filtering strategies

- **Original.** Raw LLM output, no Judge intervention. Maximum recall, most noise.
- **Fallback.** Judge’s revision applied where valid; falls back to the original if the revision is malformed. Balanced.
- **Strict.** Only triplets explicitly validated or corrected by the Judge kept. Maximum precision, smallest graph.

### 6.4 Cross-model results (Summer 2026 benchmark)

Gemini 2.5 Flash on the 127-article shared slice:

| Dimension         | Mean | Median | Std  | Min  | Max  |
|-------------------|------|--------|------|------|------|
| Coverage          | 0.74 | 0.70   | 0.07 | 0.60 | 0.90 |
| Uniqueness        | 0.83 | 0.80   | 0.10 | 0.60 | 1.00 |
| Semantic Validity | 0.84 | 0.85   | 0.07 | 0.60 | 1.00 |
| Consistency       | 0.92 | 0.90   | 0.08 | 0.70 | 1.00 |

Full comparison table (Claude / GPT / Qwen / Gemini side-by-side) — see Pierre’s benchmark deck, EIB Report 4.

## 7 Stage 3 — Knowledge graph construction

### 7.1 From triplets to graph

- Nodes: distinct entities across all triplets in a given time window.
- Node attributes: entity type (company, sector, ticker, ...).
- Edges: directed, from  $e_1$  to  $e_2$ . Attributes carry the verb and its category (SSI / GMM / GFMK).
- Edge weight: multiplicity — how many articles support this exact relationship in the window.

### 7.2 Temporal snapshots

- One graph per calendar month — 24 snapshots for the 2019–2020 corpus.
- Each snapshot is a self-contained slice: nodes and edges only from articles dated that month.
- Design rationale: financial relationships are non-stationary — what mattered in January 2020 is not what matters in April.

### 7.3 Text vs. summary — Spring 2026 finding

- Two graphs built from the same articles, one from the article body and one from an LLM-generated summary.
- *Coverage* favoured full text substantially (0.778 vs. 0.687).
- *Uniqueness* favoured the summary (0.918 vs. 0.874) — summaries are naturally less redundant.

- Downstream GAT link prediction favoured the full-text graph consistently by  $\approx 0.08$  MRR across all 10 model configurations.
- Conclusion: breadth beats precision for graph learning. Redundancy in the source is absorbed by the model; missing information cannot be.

## 7.4 Static layout (Summer 2026)

- Node positions precomputed with `networkx.spring_layout` at snapshot build time.
- Layout is deterministic — the same month always renders in the same arrangement.
- Frontend consumes these positions directly (Cytoscape “preset” layout) so the graph appears at its final view instantly, no settling animation, no drift.

# 8 Stage 4 — Graph Attention Network

## 8.1 What the model is doing, in plain English

- Assign every entity a 256-dimensional vector (its “meaning”).
- Update each entity’s vector by attending to its neighbours in the graph, weighted by how relevant each neighbour looks.
- After training, similar vectors indicate the model thinks two entities relate to each other.
- “Link prediction” = ask the model: for entity  $A$ , which unseen entities  $B$  does the model score highest? These are its predictions.

## 8.2 Architecture

| Component       | Configuration                                                                |
|-----------------|------------------------------------------------------------------------------|
| Input linear    | $33 \rightarrow 256$ (baseline) or $38 \rightarrow 256$ (causal)             |
| Node-type embed | 256-dim, one per entity type                                                 |
| Node-id embed   | 256-dim, 0.5 dropout during training                                         |
| LayerNorm       | After summing input + type + id                                              |
| GATConv layer 1 | $256 \rightarrow 256$ , 4 heads concatenated, edge-dim 33                    |
| GATConv layer 2 | $1024 \rightarrow 256$ , 1 head, mean aggregation                            |
| Edge features   | $\text{rel\_emb}(16) + \text{cat\_emb}(16) + \log(1 + w)(1) = 33\text{-dim}$ |

## 8.3 Loss functions compared

- **BPR** (Bayesian Personalised Ranking) — pairwise ranking: pull true edges above sampled negatives.

$$\mathcal{L}_{\text{BPR}} = -\frac{1}{N} \sum \ln \sigma(s(u, v^+) - s(u, v^-))$$

- **InfoNCE** (contrastive) — treat link prediction as a multi-class problem across  $K$  negatives.

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(s(u, v^+)/\tau)}{\exp(s(u, v^+)/\tau) + \sum_i \exp(s(u, v_i^-)/\tau)}$$

- **CE** (Cross-Entropy) — absolute link-existence probability.

$$\mathcal{L}_{\text{CE}} = -[y \log p + (1 - y) \log(1 - p)]$$

- **FindKG** — the margin-based loss from the FinDKG paper.
- **Hybrid** — CE + InfoNCE weighted combination.

#### 8.4 Rolling-window evaluation

- Train on months  $[t - W, t]$ , evaluate on month  $t + 1$ , roll forward, repeat.
- For each true edge in the test month, sample 50 negatives; measure whether the model ranks the true one above the negatives.
- Metrics: MRR (Mean Reciprocal Rank), Hits@1, Hits@3, Hits@10.
- “Seen” vs. “Unseen” MRR reported separately — edges between known-in-train entities vs. novel entities.

#### 8.5 Spring 2026 data-leakage fix

- Original bug: global entity index built from the entire dataset before splitting, so the embedding table implicitly contained future entities.
- Fix: build index only from the initial training window; update incrementally as the window rolls forward.
- Effect: honest evaluation — previously optimistic MRRs came down; comparisons across window sizes and loss functions became apples-to-apples.

#### 8.6 Spring 2026 window-size ablation

| Config              | Splits | MRR           | H@1    | H@3    | H@10   |
|---------------------|--------|---------------|--------|--------|--------|
| 3M train / 1M test  | 46     | <b>0.8126</b> | 0.7274 | 0.8740 | 0.9718 |
| 6M train / 1M test  | 43     | 0.7934        | 0.6981 | 0.8561 | 0.9578 |
| 6M train / 3M test  | 40     | 0.7891        | 0.6943 | 0.8512 | 0.9534 |
| 12M train / 1M test | 37     | 0.7666        | 0.6727 | 0.8275 | 0.9483 |

- Monotonic decline with longer training windows — recency wins.
- Interpretation: financial knowledge graph structure is non-stationary; older windows drag in patterns from stale regimes.
- Practical implication: a 3-month rolling window costs less compute AND performs best.

#### 8.7 Spring 2026 causal augmentation

- Baseline node features (33-dim): 32-dim name hash + 1-dim log-degree.
- Causal add-on (5-dim): in-degree, out-degree, in-edge-weight-sum, out-edge-weight-sum, PageRank — all computed on a DAG learned via NOTEARS.

- NOTEARS: continuous optimisation for learning directed acyclic graphs from observational data (Zheng et al., 2018).
- Applied to the top-200 most-mentioned entities to keep the co-occurrence matrix tractable.
- Hypothesis: causal roles are more stable than correlation patterns across market regimes.

## 8.8 Causal augmentation results

- **Covid-19 (2019–2020).** Baseline MRR fell 0.0763 from pre-shock to shock. Causal model fell only 0.0031 — a 96% reduction in shock drop.
- **Ukraine war (2021–2022).** Causal model absolute performance higher across all phases (pre-MRR 0.7290 vs. 0.6853). Within-shock standard deviation drops by  $\approx 6\times$  (MRR  $\sigma$  0.0026 vs. 0.0152).
- **Trade-off.** Causal features add *noise* during stable periods (small MRR sacrifice) in exchange for shock resilience.
- **Interpretation.** Causal features help most when the disruption is preceded by detectable shifts in news coverage (Covid); help less for sudden exogenous shocks with no prior signal (war).

## 8.9 Summer 2026 rerun

- Retrained the full 10-configuration leaderboard on the 2019–2020 corpus using the Spring 2026 rolling-window fix.
- Ran on CPU (Apple MPS backend crashed with a known Metal assertion on this graph size).
- Wall time:  $\approx 19$  minutes for the full sweep (10 configs  $\times$  23 validation months).
- Winner: **CE (No Edge Feats), MRR 0.7746.**
- Full leaderboard:

| Rank | Model                   | MRR           | H@1    | H@3    | H@10   |
|------|-------------------------|---------------|--------|--------|--------|
| 1    | CE (No Edge Feats)      | <b>0.7746</b> | 0.6891 | 0.8235 | 0.9463 |
| 2    | BPR (No Edge Feats)     | 0.7465        | 0.6596 | 0.7938 | 0.9182 |
| 3    | InfoNCE (Rel+Cat+NormW) | 0.7455        | 0.6662 | 0.7869 | 0.8988 |
| 4    | InfoNCE (No Edge Feats) | 0.7392        | 0.6595 | 0.7768 | 0.8969 |
| 5    | Hybrid (No Edge Feats)  | 0.7334        | 0.6441 | 0.7828 | 0.9092 |
| 6    | CE (Rel+Cat+NormW)      | 0.7257        | 0.6357 | 0.7738 | 0.9058 |
| 7    | BPR (Rel+Cat+NormW)     | 0.7194        | 0.6302 | 0.7637 | 0.9006 |
| 8    | FindKG (No Edge Feats)  | 0.7167        | 0.6184 | 0.7716 | 0.9135 |
| 9    | FindKG (Rel+Cat+NormW)  | 0.7049        | 0.6054 | 0.7622 | 0.9077 |
| 10   | Hybrid (Rel+Cat+NormW)  | 0.7036        | 0.6083 | 0.7575 | 0.8907 |

- Interpretation of MRR 0.77: on average the true target entity is ranked  $\approx 1.3$  out of 51 candidates. Random baseline would be  $\approx 0.05$ .
- Edge features (relationship type + category) do not consistently help — graph topology alone carries most of the signal.

- Weights saved per (loss function, edge-feature setting, month) — 230 checkpoint files.

## 9 Predictions layer (Summer 2026)

### 9.1 Design

- Modelled on the FinDKG paper’s “Top KG Entities” leaderboard (Ago et al., 2024).
- One row per top entity, per month, per column of information — see Section 9.2.
- Backed by the actual winning GAT weights (CE No-Edge-Feats), not synthetic.
- Exposed in the UI as a full-height tab alongside the graph view.

### 9.2 Columns

- **Top KG Entity.** The most influential entity in the month’s graph, per the GAT.
- **Entity Type.** Company / sector / ticker / instrument / indicator (colour-coded chip).
- **Rank Percentile.** 0–100. Position of this entity when all entities are sorted by mean outgoing GAT prediction score.
- **Novelty (z-score).** Standardised measure of how spiking this entity’s activity is versus its own historical baseline. Positive = spiking; near-zero = business as usual; negative = quieting.

$$\text{novelty}(e) = \frac{r(e) - \mu_r}{\sigma_r}, \quad r(e) = \text{edges}_{\text{recent } 3\text{mo}}(e) - \overline{\text{edges}_{\text{prior}}}(e)$$

- **Recent 3-mo Trend.** Sparkline of edge counts in months  $[P - 2, P - 1, P]$ .
- **Predicted Most Impacted Financial Entities.** Top 5 targets by GAT score, filtered to financial entity types.

### 9.3 Pipeline

- For each validation month with a GAT checkpoint:
  1. Rebuild the month’s graph.
  2. Load the CE No-Edge-Feats weights for that month.
  3. Run inference: produce 256-dim embedding per node.
  4. All-pairs dot product  $\rightarrow$  score matrix.
  5. Rank entities by mean outgoing score; compute percentile, novelty, trend, impacted list.
- All 23 monthly leaderboards baked into a single 446 KB static JSON at build time.
- UI fetches once, caches in-memory, and re-renders per selected month with zero backend calls.

## 9.4 Reality check on the data

- Nothing is LLM-generated for these leaderboards. Trend and novelty are recomputed directly from the source CSV; ranking and impacted-entity lists come from real PyTorch weight files ( $\approx$  10 MB each).
- Verified: for Texas Instruments in March 2020, novelty +3.05 and trend [13, 12, 2] both reproduce exactly from the source data.
- Failure modes visible in the output: extraction noise (stray numeric entities), duplicate entities (e.g. “TXN” and “Texas Instruments Inc” treated as separate nodes). These are upstream extraction issues, not prediction-layer issues.

## 10 User interface (Summer 2026)

### 10.1 Design goals

- Non-technical analysts should be able to browse the knowledge graph without training.
- Every claim in the UI must be traceable to a source (article or model output), not a black-box statement.
- Deployable as a static site so it runs from Vercel’s CDN with no backend infrastructure to maintain.
- Aesthetically restrained — optimised for reading in a briefing setting.

### 10.2 Architecture

- **Frontend:** Next.js 16 (App Router) + TypeScript, Cytoscape.js for graph rendering, Tailwind CSS for styling, shaden/ui for primitives, next-themes for light/dark.
- **Data layer:** static JSON files under /data/ (index, per-month snapshots, per-month article corpora, GAT predictions).
- **Optional backend:** FastAPI + Python for local pipeline re-runs and chat; not required in production.
- **Hosting:** Vercel personal account, auto-deploy on push to main. Domain: `eib-knowledge-graph.vercel.app`.

### 10.3 Features

- **Interactive knowledge graph.** Pan, zoom, hover for edge tooltips, click a node to focus its neighbourhood (dim the rest).
- **Filter rail.** Toggle entity types on/off; toggle relationship categories; minimum-connections slider (proportional to the current view’s max degree).
- **Time slider.** Dual-thumb date range. Preset chips: 1M / 3M / 6M / 12M / All. Defaults to trailing 6 months. Timeline slider drives BOTH the graph tab and the predictions tab.
- **Safari-style tabs.** Knowledge graph / Predictions. Only one visible at a time; state preserved on switch (with a Cytoscape lifecycle fix for the fit-viewport race condition).

- **Predictions view.** FinDKG-style table as described in Section 9.
- **Entity search (CK).** Command-palette-style entity picker across the current snapshot.
- **Article search.** Keyword search across the source news corpus (per-month JSON bundles fetched in parallel, scored client-side).
- **Chat panel.** Optional LLM chat over the corpus (backend-only; user supplies their own API key).
- **Physics settings.** Optional d3-force live simulation for the graph, behind a gear icon. Off by default — static precomputed positions are the default view.
- **Theme.** Light mode by default; dark mode toggle in the header.

## 10.4 Recent UI iterations (Summer 2026, chronological)

- Replaced polarity edge colouring with 8-family causal-type colouring (Okabe-Ito colourblind-safe palette).
- Fixed graph layout to precomputed `spring_layout` positions — no more settle-from-chaos on page load.
- Added dual-thumb date range selector with client-side snapshot merging.
- Added Safari-style tab strip; predictions moved from a bottom drawer into a full tab.
- Removed purple / neon accents; migrated to neutral palette.
- Fixed Cytoscape crash on tab switch: cancelled pending `requestAnimationFrame` on unmount.

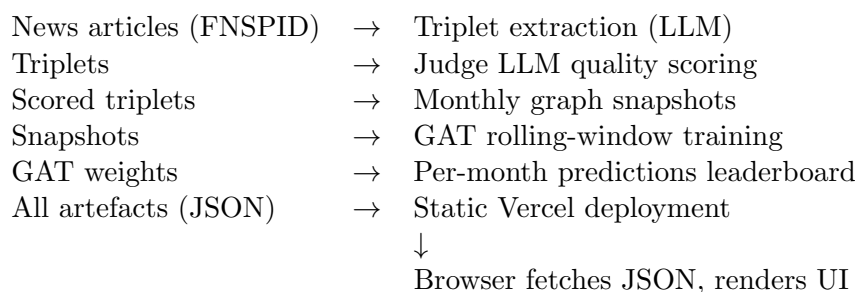
## 10.5 Late-Summer 2026 overhaul

- **Multi-source architecture.** A source catalogue drives a data-source dropdown; each source declares which tabs it supports (FNSPID: graph + predictions; STOXX: graph + factor analysis; VM run: graph). Unsupported tabs are hidden, not greyed. Per-source data lives under `/data/sources/{id}/`.
- **Timeline redesign.** Months are equal cells of the track (a 1-month selection is a full cell, not a point). The selected range is a visible window block: drag the middle to slide through time (glides with the pointer, snaps to month boundaries), drag the edges to resize, click the track to jump; preset chips 1M/3M/6M/12M/All; default trailing quarter. On the Predictions tab the same track shows the training months (amber) vs. the predicted month (emerald) of the rolling-window GAT.
- **Obsidian-style graph layout.** A real d3-force simulation runs to convergence before render: many-body repulsion, degree-scaled link springs, a collision force guaranteeing zero node overlap, and centring gravity. Deterministic (seeded), so a given view always lays out identically.
- **Readability at scale.** Node and label sizes are zoom-compensated to constant pixel size across corpora; node size normalised to each snapshot’s own degree ceiling; permanent labels allocated by a collision-avoiding budget (all names in moderate or multi-cluster views, top-40% in dense

single-blob views, every cluster’s top entity always named); auto density threshold targets the  $\approx 300$  most-connected nodes rather than a fixed percentage of the maximum degree.

- **Industry filter (STOXX).** A sector section above the entity-type checkboxes. Sectors are inferred per news cluster from the watchlist company it concerns; default view opens on Pharma with the other five sectors one click away.
- **Factor analysis tab.** As described in Section 13.
- **Dead-control hygiene.** Controls that do not affect a given tab are hidden there (Factor analysis shows neither the filter rail nor the timeline).
- **Report tab.** This report, viewable in-app as an embedded PDF.

## 10.6 Data flow diagram



## 11 Live corpus — STOXX Europe 600 pipeline (Summer 2026)

### 11.1 Motivation

- Everything upstream operated on a fixed historical corpus. Question: can the whole chain — fetch, extract, build graph, deploy — run *live*, unattended?
- Chosen universe: STOXX Europe 600 subset. European coverage is directly sponsor-relevant (EIB).
- Scope deliberately small: 30 curated tickers across six sectors (pharma, tech, banks, industrials & autos, energy, consumer & luxury) so a full daily run fits comfortably in a GitHub Actions job.

### 11.2 Daily pipeline (fully automated)

- GitHub Actions cron, 07:30 UTC daily. No human in the loop; verified self-healing over multiple days (auto-commit with pull-rebase retry when other pushes land mid-run).
- Steps: fetch Google News RSS per ticker → drop stale republished articles (>120 days old — these previously spawned junk one-article months as far back as 2019) → dedupe against a URL ledger → Gemini 2.5 Flash extraction → refinement passes (Section 12) → merge into monthly snapshots → recompute factor bundle → commit. The commit triggers the Vercel redeploy.
- **Enriched extraction schema.** Each triplet carries three extra fields beyond subject/relation/object: sentiment  $[-1, 1]$ , materiality (US-dollar magnitude mentioned with the fact), and an event-type tag. These feed the factor model.

- **Raw triplet persistence.** Every article’s exact triplet list is stored per month before aggregation (`triplets/YYYY-MM.json`) — article-level provenance, exact factor recomputation, and a future retraining dataset.
- Index hygiene: months whose snapshot holds fewer than 3 edges are excluded from the timeline (stray misdated articles cannot pollute the axis).
- Cost: on the order of cents per day (Gemini 2.5 Flash over  $\approx 90$  title-length articles). Runs until the API key is retired; the site then simply stops updating and keeps serving the last state.
- Known limitation: Google News RSS supplies titles only ( $\approx 60$  characters of signal per article) and no deep history — the corpus accumulates forward from activation.

## 12 Extraction refinement — three-LLM parity (Summer 2026)

- The team pipeline (Fall 2025 onward) is a **three-LLM design**: (1) a triplet-extraction LLM; (2) a Judge LLM scoring each article’s triplets on coverage, uniqueness and semantic validity; (3) a *flagged-triplet refinement* LLM that receives the article, the bad triplets and the judge’s verdict, and regenerates better ones.
- The live pipeline initially ran stage 1 only (plus regex filters) — the source of duplicate entities (“ASML” / “ASML Holding” / “ASML Holding N.V.”) and nonsense nodes (bare numbers, generic phrases).
- Now added, adapted to the title-only corpus and collapsed for cost:
  - **Judge-and-refine pass** (one Gemini call per article). Drops entities that are meaningless as graph nodes, merges restated facts, normalises names within the article.
  - **Cross-article entity canonicalisation** (one call per run). Maps name variants onto one canonical name each, preferring names already in the corpus so new edges merge into existing nodes instead of duplicating them. Tickers are never merged into companies.
- Both passes *fail open*: an API or parse failure keeps the first-pass triplets, so a flaky call degrades quality for a day rather than losing data.

## 13 News factor model (Summer 2026)

### 13.1 Idea

- First-principles question: beyond drawing a graph, what is the *analytical* utility of daily triplets? Answer: treat news coverage itself as a measurable signal — a Fama–French-style factor decomposition of the news, per entity.
- Every company/topic in the corpus is scored on five factors, i.e. five plain questions about its recent coverage:
  - **Attention** — how much is it written about?
  - **Sentiment** — is the tone positive or negative?

- **Consensus** — do sources agree, or is the story contested?
- **Novelty** — is it generating its own storylines or only appearing in others’?
- **Materiality** — how much money do its stories involve (deal sizes, fines, revenue figures)?

## 13.2 Method

- Factors roll up from the last two months of stored snapshots (a rolling corpus), not just the day’s new articles — the daily delta after ledger warm-up is a handful of articles and produced empty bundles until this fix.
- Scores standardised to z-scores; **KMeans** ( $k = 5$ ) clusters entities in the full five-factor space — no sector labels or company information supplied. Clusters read as automatically discovered *news archetypes*.
- Observed clusters are economically sensible: e.g. Novo Nordisk / Shell / Volkswagen form the high-attention–low-consensus group (heavily covered, contested stories); an energy-heavy cluster (BP, Eni) carries the high-materiality–positive-sentiment signature.
- **PCA** projects the five scores to two axes for display only ( $\approx 76\%$  of variance in three components); clustering happens in the full space.
- UI: dedicated “Factor analysis” tab — labelled scatter (collision-avoided names, zoom/pan with constant-size labels so overlaps separate on zoom), archetype cluster cards with plain-language signatures, and a collapsible analyst explainer covering what the tab is for, the factors, the clusters and the map, in non-engineer language.

## 14 VM-to-UI publishing pipeline (Summer 2026)

- The team’s production extraction runs on a Google Cloud VM (`eib-central1a`; Ollama + Qwen 2.5 14B; `tmux-detached` runs over the full *FNSPID* corpus). Outputs accumulated on the VM with no path to the UI.
- Built `scripts/publish_vm_output.py`: converts the pipeline’s output CSVs (article rows with 6-tuple triplets) into the site’s data format — monthly snapshots with layout positions, article records, raw triplet provenance — and pushes; the push triggers the Vercel redeploy. Safe re-publish semantics (`--replace`), dry-run mode, tolerant CSV parsing for large files.
- **First production use.** The full 2019–2021 extraction (Qwen 2.5 14B, pushed to the team pipeline repository on 16 Aug) was published as a third UI source: *FNSPID — qwen2.5 VM run (2019–2021)*. 5,358 articles, 30,861 triplets, all 25 months browsable.
- Kept separate from the original *FNSPID* source so the GAT predictions remain consistent with the extraction they were trained on.
- Documented one-time VM setup (write-scoped deploy key + one cron line) in `docs/VM_PUBLISHING.md`; alternatively a repository-side mirror workflow can poll the public pipeline repo for new outputs, requiring nothing from the VM at all.

## 15 Deployment and infrastructure

- **Repository.** <https://github.com/ap4509-columbia/eib-knowledge-graph> (team collaborators: SigmaWave, Ruiwen04).
- **Vercel project.** eib-knowledge-graph. Auto-deploy on push to main. Aliased domains: eib-knowledge-graph.vercel.app + Git-branch preview URLs.
- **Static-only production build.** No backend at runtime. All data ships as JSON files served from the CDN.
- **Self-updating scraper** (live in production for the STOXX corpus — Section 11).
  - Sources: Google News RSS (per-corpus watchlists), ECB / ESMA / BoE regulator RSS feeds.
  - Per-corpus watchlists: US semiconductors, EU utilities, EU banks, EU regulators, STOXX Europe 600 (30-ticker curated subset in production).
  - Deduplication ledger prevents re-processing.
  - Runs daily as a GitHub Actions cron and auto-commits new snapshots; each commit triggers the Vercel redeploy.
- **Local backend** (optional). FastAPI + uvicorn on port 8000 for pipeline re-runs and chat during development.

## 16 Team contributions summary

| Team / semester | Members                        | Key contributions                                                                                                                                                                                                                                                           |
|-----------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fall 2025       | Lee, Gao, Lu, Cuadros          | <ul style="list-style-type: none"><li>• Baseline end-to-end pipeline</li><li>• Triplet extraction prompt + ontology</li><li>• Judge LLM design (4 dimensions, 3 strategies)</li><li>• First GAT for link prediction</li><li>• Basic HTML visualisation</li></ul>            |
| Spring 2026     | Xixi Chen, Xinyi Chen, Yi Yang | <ul style="list-style-type: none"><li>• Full-text vs. summary extraction study</li><li>• Rolling-window data-leakage fix</li><li>• 3M / 6M / 12M window ablation</li><li>• Causal feature augmentation via NOTEARS</li><li>• Covid-19 + Ukraine war event studies</li></ul> |

Summer 2026

Alexandra Paiz, Pierre Pujol, Ruiwen Wang

- Multi-model LLM benchmark (Claude / GPT / Qwen / Gemini)
  - Interactive web application (Next.js + Cytoscape)
  - FinDKG-style predictions leaderboard
  - GAT retraining with fixed rolling window (MRR 0.7746)
  - Self-updating scraper framework
  - Vercel deployment
- 

## 17 Deliverables

### 17.1 Fall 2025

- `summary_triplets_metrics_22.csv` — earlier scored triplet dump.
- Original pipeline modules under `components/`.
- Static per-month HTML visualisations.

### 17.2 Spring 2026

- Written report: *LLM and Graphs for Financial Texts* (EIB1\_Report.pdf, 16 pages).
- Poster.
- `summary_triplets_19_20.csv` — the 2-year triplet corpus this project has used ever since.
- Corrected rolling-window code (`Rolling Window/` folder).
- NOTEARS causal feature pipeline (`causal_files/` folder).

### 17.3 Summer 2026

- Deployed web application: <https://eib-knowledge-graph.vercel.app>.
- Source repository (private).
- Multi-LLM benchmark: 4-model comparison + Judge LLM stats.
- Retrained GAT: 230 checkpoints + leaderboard CSV.
- Predictions JSON: 23 monthly leaderboards, 446 KB.
- Self-updating scraper framework + per-corpus watchlists.
- This report.

## 18 Future work

### 18.1 Corpus extension

- Currently 2 years of triplets (2019–2020). The underlying FNSPID dataset extends 1999–2023 with  $\approx 15\text{M}$  articles.
- Extending to 5 years:  $\approx \$30$  in LLM extraction cost, 2–3 days of processing.
- Extending to full 15 years:  $\approx \$200$ .
- Blocker: team budget decision, pending.

### 18.2 Cross-industry expansion

- Current focus: semiconductors.
- Watchlists already defined for EU utilities, EU banks, EU regulators, STOXX Europe 600.
- Enables sponsor-relevant coverage (European infrastructure, regulation, banking).

### 18.3 Sponsor-facing improvements

- Entity aliasing (dedupe “TXN” and “Texas Instruments Inc”) — *shipped for the live corpus* via the canonicalisation pass (Section 12); extending it to the historical FNSPID sources is a one-off batch job over the now-persisted raw triplets.
- Currency numeral cleanup (bare numbers like “1840” as CURRENCY entities) — *shipped for the live corpus* via hard filters + the refine pass; pending for historical sources.
- Promote unmodelled entity types (PERSON, COUNTRY, LOCATION, INSTITUTION) to first-class schema types.
- Sentiment scores from Marketaux or equivalent as a second signal alongside triplets.

### 18.4 Model integration

- Per-edge GAT confidence exposed as an “uncertainty” colour mode on the graph.
- Similar-entity retrieval from the node embeddings in the detail panel.
- Predicted *new* edges (not just ranking known ones) drawn on the graph with a distinct style.

### 18.5 Additional event studies

- Interest-rate cycles.
- European energy price spike (2022–2023).
- Banking-sector stress episodes.

## A Metric definitions — quick reference

- **MRR** (Mean Reciprocal Rank).  $MRR = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q}$ . Where  $\text{rank}_q$  is the position of the true target for query  $q$ .
- **Hits@k**. Fraction of queries where the true target is in the top  $k$ .
- **Coverage**. Fraction of salient article content captured by the extracted triplets.
- **Uniqueness**. 1– redundancy rate among the triplets (cosine similarity in embedding space).
- **Semantic Validity**. Fraction of triplets that fit the financial ontology.
- **Consistency**. Absence of pairwise contradictions.
- **Novelty z-score**. Standardised measure of an entity’s activity spike vs. its own history.
- **Rank Percentile**. Position of an entity when ranked by mean outgoing GAT prediction score, expressed as a percentile.

## B Software stack — full list

- **Extraction**: Python, Ollama (Qwen 2.5 14B / 32B), Google Gemini API (google-genai), Anthropic API, OpenAI API.
- **Modelling**: PyTorch, PyTorch Geometric, NetworkX, SciPy, scikit-learn (PCA, KMeans), SentenceTransformers.
- **Backend**: FastAPI, uvicorn, pandas.
- **Frontend**: Next.js 16, TypeScript, Cytoscape.js, cytoscape-d3-force, Tailwind CSS, shadcn/ui, next-themes, lucide-react.
- **Infrastructure**: GitHub, GitHub Actions (cron), Vercel (deployment, CDN).

## References

- [1] Ago, T. et al. (2024). *FinDKG: Dynamic Knowledge Graphs with Large Language Models for Detecting Global Trends in Financial Markets*. <https://arxiv.org/abs/2407.10909>
- [2] Zheng, X., Aragam, B., Ravikumar, P., & Xing, E. P. (2018). *DAGs with NO TEARS: Continuous optimization for structure learning*. Advances in Neural Information Processing Systems, 31.
- [3] Lee, G., Gao, J., Lu, X., & Cuadros, A. (Fall 2025). *LLM and Graphs for Financial Texts* (baseline pipeline report).
- [4] Chen, Xixi; Chen, Xinyi; Yang, Yi (Spring 2026). *LLM and Graphs for Financial Texts* (Spring 2026 EIB1 report).
- [5] Zihan, D. et al. FNSPID: Financial News and Stock Price Integration Dataset. Hugging Face: <https://huggingface.co/datasets/Zihan1004/FNSPID>